

A Full Solution of the Real Beta Equipoint Conjecture

Run `fa_banach_001`, agent `agent_lane_11`

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Status. Candidate full solution, likely valid, pending human review. This packet proves Conjecture 1.13 of Helton–Klep–McCullough–Schweighofer; it does not address their separate Conjecture 1.16.

1 Source problem

For $a, b > 0$, write $X_{a,b}$ for a beta random variable, with density

$$\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}, \quad 0 < x < 1.$$

The source defines an equipoint $e_{a,b}$ and states that its inequality $e_{a,b} \leq a/(a+b)$ is equivalent to the following comparison of beta tails. Conjecture 1.13 on printed and PDF page 13 asks whether the comparison holds for all real $a \geq b > 0$ [1].

The next two subsections contain more information on equipoints and how they compare to means and medians.

1.8.3. *Equipoints compared to medians.* Here is a basic property of equipoints versus medians and means. Let $\frac{1}{2}\mathbb{N}$ denote the set of all positive half-integers, i.e., all $\mathfrak{s} = \frac{s}{2}$ with $s \in \mathbb{N}$.

Theorem 1.11. *For $\mathfrak{d} \in \frac{1}{2}\mathbb{N}$ and $\mathfrak{s}, \mathfrak{t} \in \frac{1}{2}\mathbb{N}$ with $\mathfrak{s} + \mathfrak{t} = \mathfrak{d}$, if $\frac{\mathfrak{d}}{2} \leq s < \mathfrak{d}$, then*

$$(1.15) \quad P_{\frac{\mathfrak{s}}{\mathfrak{d}}}(\mathfrak{S} < \mathfrak{s}) \leq P_{\frac{\mathfrak{s}}{\mathfrak{d}}}(\mathfrak{S} > \mathfrak{s}) \quad \text{provided } \mathfrak{s}, \mathfrak{t}, \mathfrak{d} \in \mathbb{N};$$

$$(1.16) \quad P^{b(\mathfrak{s}, \mathfrak{t}+1)}\left(\mathfrak{B} \geq \frac{\mathfrak{s}}{\mathfrak{d}}\right) \leq P^{b(\mathfrak{s}+1, \mathfrak{t})}\left(\mathfrak{B} \leq \frac{\mathfrak{s}}{\mathfrak{d}}\right).$$

Both (1.15) and (1.16) are equivalent to

$$(1.17) \quad e_{\mathfrak{s}, \mathfrak{t}} \leq \frac{\mathfrak{s}}{\mathfrak{d}}.$$

We also have the lower bound

$$(1.18) \quad \frac{\mathfrak{s} + 1}{\mathfrak{s} + \mathfrak{t} + 2} \leq e_{\mathfrak{s}, \mathfrak{t}},$$

for real numbers $\mathfrak{s} \geq \mathfrak{t} \geq 1$.

Remark 1.12. The inequality (1.16) for integer $\mathfrak{s}, \mathfrak{t}$ is Simmons' Theorem, cf. [PR07]. The lower bound is new. For half-integer $\mathfrak{s}, \mathfrak{t}$ both our upper and lower bounds are new. $\square$

Proof. The inequality (1.16) implies (1.15) by (1.10) and (1.11). However, (1.16) and $e_{\mathfrak{s}, \mathfrak{t}} \leq \frac{\mathfrak{s}}{\mathfrak{d}}$ is the content of Theorem 10.1. The lower bound (1.18) is Proposition 11.2. $\blacksquare$

Computer experiments lead us to believe (1.17) is true for real numbers:

Conjecture 1.13. For $\mathfrak{s}, \mathfrak{t} \in \mathbb{R}_{>0}$ with $\mathfrak{s} \geq \mathfrak{t}$, inequality (1.16) holds. Equivalently, $e_{\mathfrak{s}, \mathfrak{t}} \leq \frac{\mathfrak{s}}{\mathfrak{s} + \mathfrak{t}}$.

As a side-product of our quest for bounds on the equipoint we obtain new upper bounds on the median $m_{\alpha, \beta}$ of the Beta Distribution $\text{Beta}(\alpha, \beta)$.

Transcription. For $a, b \in \mathbb{R}_{>0}$ with $a \geq b$, prove

$$\mathbb{P}\left(X_{a, b+1} \geq \frac{a}{a+b}\right) \leq \mathbb{P}\left(X_{a+1, b} \leq \frac{a}{a+b}\right). \quad (1)$$

Equivalently, prove $e_{a, b} \leq a/(a+b)$.

2 Proof intuition

The two sides of (1) initially look like unrelated incomplete-beta tails. Passing to odds, $z = x/(1-x)$, places both tails on the beta-prime line, separated at $q = a/b$. Scaling the lower half by $z = qu$ and inverting the upper half by $z = q/u$ puts both integrals on $(0, 1)$.

The resulting integrand difference has one sign change: it is negative near zero and positive afterward. There is also an exact weighted cancellation with weight $(1-u)/u$. That cancellation is the zero first centered moment of a beta-prime density whose mean is q . Because the weight is strictly decreasing, cancellation under that weight can occur only if the unweighted positive area is larger than the unweighted negative area. This is exactly the desired inequality.

3 The theorem

Theorem 1. *For every $a \geq b > 0$, inequality (1) holds. Equality holds when $a = b$, and the inequality is strict when $a > b$. Consequently,*

$$e_{a,b} \leq \frac{a}{a+b} \quad (a \geq b > 0),$$

with equality exactly in the symmetric case $a = b$.

4 Proof

Put

$$d = a + b, \quad m = \frac{a}{d}, \quad q = \frac{a}{b} = \frac{m}{1-m}.$$

The beta recurrences

$$B(a+1, b) = \frac{a}{d} B(a, b), \quad B(a, b+1) = \frac{b}{d} B(a, b)$$

show that the right side of (1) minus the left side has the sign of

$$\frac{1}{a} \int_0^m x^a (1-x)^{b-1} dx - \frac{1}{b} \int_m^1 x^{a-1} (1-x)^b dx. \quad (2)$$

With the odds substitution $z = x/(1-x)$, expression (2) becomes

$$\frac{1}{a} \int_0^q \frac{z^a}{(1+z)^{d+1}} dz - \frac{1}{b} \int_q^\infty \frac{z^{a-1}}{(1+z)^{d+1}} dz. \quad (3)$$

In the first integral set $z = qu$; in the second set $z = q/u$. Since $q/a = 1/b$, the two resulting positive prefactors agree. Therefore (3) has the sign of

$$H := \int_0^1 (P(u) - Q(u)) du, \quad (4)$$

where

$$P(u) = \frac{u^a}{(1+qu)^{d+1}}, \quad Q(u) = \frac{u^b}{(q+u)^{d+1}}. \quad (5)$$

We first prove an exact weighted cancellation. Let $w(u) = (1-u)/u$. Undoing the two substitutions above gives

$$\begin{aligned} \frac{q^a}{b} \int_0^1 w(u) P(u) du &= \frac{1}{a} \int_0^q (q-z) \frac{z^{a-1}}{(1+z)^{d+1}} dz, \\ \frac{q^a}{b} \int_0^1 w(u) Q(u) du &= \frac{1}{a} \int_q^\infty (z-q) \frac{z^{a-1}}{(1+z)^{d+1}} dz. \end{aligned}$$

The density kernel

$$\varphi(z) = \frac{z^{a-1}}{(1+z)^{d+1}}$$

has mean q after normalization, because

$$\frac{\int_0^\infty z \varphi(z) dz}{\int_0^\infty \varphi(z) dz} = \frac{B(a+1, b)}{B(a, b+1)} = \frac{a}{b} = q.$$

Its lower and upper first deviations about q are consequently equal. Thus

$$\int_0^1 w(u)(P(u) - Q(u)) du = 0. \quad (6)$$

All integrals here are finite for $a, b > 0$; near zero the only potentially singular weighted term is of order u^{b-1} .

It remains to determine the sign pattern of $P - Q$. If $a = b$, then $q = 1$ and $P = Q$, so equality follows. Suppose $a > b$, hence $q > 1$. For $0 < u < 1$, set

$$R(u) = \frac{P(u)}{Q(u)} = u^{a-b} \left(\frac{q+u}{1+qu} \right)^{d+1}, \quad g(u) = \log R(u).$$

We have $g(u) \rightarrow -\infty$ as $u \downarrow 0$ and $g(1) = 0$. Moreover,

$$\begin{aligned} g'(u) &= (a-b) \frac{1}{u} - (d+1) \frac{q^2 - 1}{(q+u)(1+qu)} \\ &= (q-1) \left[\frac{b}{u} - \frac{(q+1)(d+1)}{(q+u)(1+qu)} \right]. \end{aligned} \quad (7)$$

The sign of the bracket is the sign of

$$N(u) = bqu^2 - (2bq + q + 1)u + bq. \quad (8)$$

The two roots of N have product one and sum

$$2 + \frac{q+1}{bq} > 2.$$

Hence exactly one root lies in $(0, 1)$ and the other lies in $(1, \infty)$. It follows that g first increases and then decreases on $(0, 1)$. Since $g(1) = 0$ and $g'(1) < 0$, g is positive immediately to the left of one. Consequently there is a unique $c \in (0, 1)$ such that

$$P(u) - Q(u) < 0 \quad (0 < u < c), \quad P(u) - Q(u) > 0 \quad (c < u < 1). \quad (9)$$

Finally, $w(u) = (1-u)/u$ is strictly decreasing. Using (6) and the crossing point c , we obtain

$$0 = \int_0^1 (w(u) - w(c))(P(u) - Q(u)) du + w(c)H.$$

By (9), the first integral is strictly negative: on each side of c , its two factors have opposite signs. Since $w(c) > 0$, this forces $H > 0$. Equations (2)–(4) now give the strict form of (1) when $a > b$. The source proves the equivalence with the equipoint formulation, completing the proof of Theorem 1. $\square$

5 Verification report

Step	Status	Verifier note
Beta normalization	valid	Both recurrence factors are positive; the desired probability difference has exactly the sign of (2).
Tail substitutions	valid	Odds, scaling, and inversion give the common factor $q^a/b > 0$ and functions (5).
Weighted identity	valid	It is equivalent to the zero centered first moment of the beta-prime kernel; integrability also holds when a parameter is below one.
One crossing	valid	The derivative numerator has reciprocal positive roots, exactly one of them in $(0, 1)$.
Final sign	valid	A decreasing weight and a negative-to-positive single crossing make the centered weighted integral strictly negative, forcing $H > 0$.
Endpoint case	valid	For $a = b$, symmetry gives $P = Q$ and $e_{a,a} = 1/2$.

Verdict: likely valid; candidate full solution. The included script `code/verify_equipoint.py` compares the original incomplete-beta gap with the transformed integrals, checks (6), verifies the crossing pattern, and runs a fixed-seed scan of 250,000 parameter pairs. No numerical contradiction was found. These computations are diagnostics, not dependencies of the proof.

6 Novelty, scope, and review recommendation

The run indexes were searched for arXiv:1412.1481, the exact conjecture label, equipoints, Simmons' theorem, and the incomplete-beta inequality. A bounded external search used the exact label and wording, source title and authors, equipoint notation, and close incomplete-beta phrases. It found the source arXiv, AMS, and author-hosted versions, plus adjacent incomplete-beta work, but no later proof, counterexample, or claimed resolution of Conjecture 1.13.

This packet resolves Conjecture 1.13 only. The distinct monotonicity Conjecture 1.16 on the next page remains outside its scope. Human review should focus on the common prefactor in (4) and the direction of the final decreasing-weight comparison.

References

- [1] J. W. Helton, I. Klep, S. McCullough, and M. Schweighofer, *Dilations, Linear Matrix Inequalities, the Matrix Cube Problem and Beta Distributions*, arXiv:1412.1481; Mem. Amer. Math. Soc. 257 (2019), no. 1232, [doi:10.1090/memo/1232](https://doi.org/10.1090/memo/1232).